ME 4224 Final Report: Turbofan Engine Design Through GasTurb 14

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Overview

Project Goal

The goal of this report is to demonstrate applicable knowledge of gas turbine engines through utilizing a simulation tool called GasTurb 14 to analyze and design a two-shaft turbofan engine and a three-shaft turbofan engine through multiple design iterations. Namely, cycle design techniques such as parametric studies, and optimizations will be performed to achieve a chosen target net thrust (FN) and a target specific fuel consumption (SFC). The target net thrust is given to be 150 KN at take-off conditions. The target SFC is chosen to be 0.4 lbm/(h*lbf) which corresponds to about 11.33 g/(KN*s). This value dictates a firm base value for the performance capability of modern turbofan engines [1].

The importance of such exercise manifests itself in an improved overall understanding of the gas turbine cycle and the corresponding relationships between relevant design parameters such as the compressor and turbine efficiencies, fan bypass ratio (BPR), overall pressure ratio (OPR), compressor pressure ratio (CPR), turbine inlet temperature (T4), engine cooling efficiency amongst many others. Beside the effect of performance parameters on one another, engineers can observe relationships between performance parameters and related geometric and material choice parameters. For example, many modern engines are being manufactured from carbon fiber composites instead of traditional metals like aluminum and titanium. This is because these composites are deemed more structurally robust and can withstand much higher temperatures at the combustor's exit which in turns improves the engine's thermal efficiency, resulting in an increased net thrust and lower SFC [1].

The design process presented in this report will be centered around four main performance parameters that will be carefully analyzed and adjusted to achieve the specified fuel consumption and the required net thrust. Those are the compressor pressure ratio (CPR), turbine inlet temperature (T4), fan bypass ratio (BPR), and the fan pressure ratio (FPR) [1]. Those parameters directly influence the thermal and propulsive efficiencies which are the main variables in lowering the SFC. Increasing the thermal efficiency stipulates the increase in both the CPR and the TIT (T4). Consequentially, improving the propulsive efficiency occurs by increasing the FPR and the BPR [1]. Other derivative parameters that could be influenced are the inlet mass flow rate and any introduced bleed air cooling.

Engine Selection

2-Spool Engine Selection

The choice of the baseline engine from which adjustments will be made to achieve subsequent design goals relied on state-of-the-art technologies available on the market that offer the best cycle efficiency ratings and the best machining and material choice selections. The GE9x, shown in Figure 1, is a high-bypass unmixed turbofan engine developed by General Electric Aviation and was chosen as the 2-Spool baseline configuration as it is one of the most efficient and powerful engines available for the current and next generation of aviation. It is designed to be coupled with the future Boeing 777X to pioneer long-haul civil flight operations. The GE9x is designed to achieve a 10% reduction in fuel consumption compared to the GE90 turbofan engine [2]. It is machined from Ceramic Matrix Composites (CMCs) which is a material combination between ceramics, namely carbon fiber and other specified composites [3]. CMCs are 2X stronger than traditional metals; they allowed the GE9x to be one-third the weight of the GE90; and they reduced cooling requirements by 20% compared to the GE90 [2]. The baseline performance parameters and the engine architecture for the GE9x are presented in the "Engine Design Process" section of the report.

Figure 1: Image of the GE9x turbofan engine

3-Spool Engine Selection

Through adequate investigation, the Rolls-Royce Trent XWB engine, shown in Figure 2, was selected as the baseline, 3-Spool engine configuration pertaining to this project. The RR Trent XWB is an unmixed high-bypass turbofan engine developed by Rolls-Royce Holdings that offers state-of-the-art performance and efficiency capabilities for modern three-spool turbofan engines. The Trent XWB is housed on the recently developed Airbus A350, which is a long-haul widebody aircraft developed by Airbus. The newest variant of the Trent XWB, the XWB-97, is optimized for the longer-range Airbus A350-1000 and future variants of the aircraft [4]. This new design was serviced in 2017 and was a step up from the previous variant the XWB-84 which powered the A350-900 [5]. Namely, the XWB-97 was made to have an increased fan spool speed by 6% from the XWB-84 variant and was redesigned to be lighter weight through the integration of specialized metal alloys, titanium, and CMCs [4]. The Trent XWB additionally offers a 15% reduction in fuel consumption in comparison to original Trent models such as the Trent 700 [6]. The baseline engine performance parameters and detailed engine architecture is presented in the "Engine Design Process" section of the report.

Figure 2: Image of the RR Trent XWB turbofan engine

Engine Design Process

Baseline Target Engine Specifications and Architecture

2-Spool Target Engine Specifications

Presented in Table 1 are the baseline engine specifications for the GE9x turbofan engine. The parameters shown in Table 1 are quoted from the book "Propulsion and Power: An Exploration of Gas Turbine by Performance Modeling" by Joachim Kurzke and Ian Halliwell, and from the literature case study done by Fanzago et al using the GE9x engine as the baseline model [1], [2]. Namely, efficiency values were taken from a table provided in [1] that provides a valid approximation of efficiency parameters for modern high-bypass turbofan engines. This is because OEMs do not disclose such data for the public domain. The remaining parameters which include pressure ratios, take-off net thrust among others were directly quoted from [2]. It is worthy to note that parameters such as the turbine inlet temperature (T4), the air mass flow rate, and the fan pressure ratio (FPR) for the GE9x engine were obtained by assumptions and referrals to sources other than [1] and [2]. Namely, the turbine inlet temperature was directly obtained from a brochure posted by GE accessible through the attached link. The air mass flow rate was not directly specified but was assumed to be 1500 kg/s since the Trent XWB with similar inlet conditions has a mass flow rate of 1440 Kg/s [8]. This flow rate is an initial value that acts as a starting point for the design process later in the report. It is expected to be changed continuously until the expected net thrust is achieved. The FPR was also not directly obtainable from the utilized sources and any online open sources. The fact that the FPR decreases as the BPR increases was taken into consideration to assume a FPR for the GE9x engine [7]. Xue et al listed the shooting range for FPRs in relation to multiple bypass ratios. The BPR for the Trent XWB was 9.3, which is close to the BPR of the GE9x. The range provided for this BPR was a FPR between 1.4 and 1.65. The quoted FPR for the Trent XWB was 1.6, therefore since the BPR of the GE9x is on the larger side, a FPR of 1.5 was assigned as seen in Table 1.

Table 1: GE9X engine baseline parameters

Design Parameter	Value (Units)	Justification
Net Thrust (FN) at **Take-Off**	110000 (lbf) / 489.3 (KN)	Aircraft Database GE9x
Inlet Air Mass Flow Rate (W)	1500 (Kg/s)	[8]
Overall Pressure Ratio (OPR)	61:1	[2]
High Pressure Compressor (HPC) Pressure Ratio	27:1	[2]
Turbine Inlet Temperature (T4)	2200 (K)	[2]
Fan Pressure Ratio (FPR)	1.5	[7], [8]
Bypass Ratio (BPR)	10:1	[2]
Fan Inner/Outer Efficiencies (Polytropic)	0.91	[1]
HPC/LPC Efficiencies (Polytropic)	0.91	[1]
LPT Efficiency (Polytropic)	0.91	[1]
HPT Efficiency (Polytropic)	0.89	[1]
Mechanical Efficiency	0.98	[2]

2-Spool Target Engine Architecture

The Ge9x engine is made through a twin-spool or two-shaft unmixed exhaust design architecture. The two-shaft aspect of this design, shown in Figure 3, means that the low-pressure compressor (LPC) stages including the fan are connected to the low-pressure turbine (LPT) stages through a shaft called the N1 shaft or the outer shaft. The remaining high-pressure compressor stages near the combustion chamber are connected to the high-pressure turbine stages through another shaft called the N2 shaft or the inner shaft. This design choice is maintained by most OEMs including GE and Pratt & Whitney (PW) as it reduces maintenance complexity, mechanical loss, and weight in comparison to three spool turbofans. The main disadvantage in the two-spool configuration manifests itself in performance. Namely, as the fan gets larger, it is ever harder to match the LPC rotation speeds with the LPT rotation speeds and this can hinder an engine's performance potential [9]. However, OEMs now are excelling in developing new geared turbofan engines, where the gear connecting the N1 shaft with the fan can better regulate LPC and fan speeds and prevent the fan tip from reaching supersonic speeds [9].

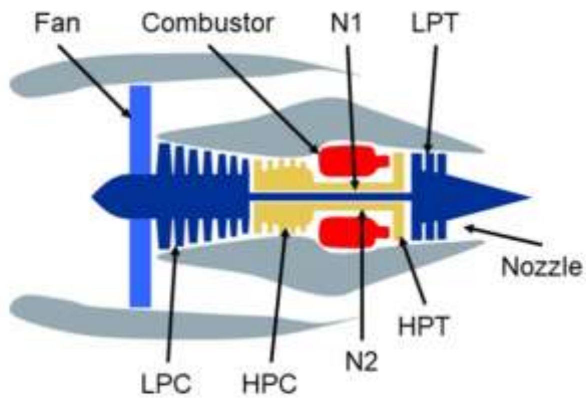

Figure 3: Simple schematic of a two-shaft turbofan engine configuration

The unmixed exhaust aspect of the GE9x engine's design architecture represents the interaction between the jet velocity of the engine's core and the engine's bypass stream [1]. Mixed turbofans are often designed to have a mixer that mixes the jet streams and improves thermal efficiency and SFC of turbofan engines [1]. However, when the bypass ratio reaches large limits, the mixer adds much more complexity and weight negating the initial benefit. This is why most modern high-bypass turbofan engines implement two nozzles for each stream at the exhaust and separate the flow.

To fully detail the GE9x engine architecture, a detailed engine schematic for a two-spool configuration through GasTurb 14 is provided in Figure 4 with specified geometrical values. It is worthy to note that the geometry of the engine in GasTurb 14 remains unchanged as the main purpose of this report is to analyze and optimize the thermodynamic cycle which is not affected notably by geometric manipulations. The main geometric parameters for the GE9x engine are compiled in Table *2* for reference.

Front LP Shaft Cone Length	m	0.168466
Middle LP Shaft Length	m	0.540032
Middle LP Shaft Radius	m	0.0098183:
Rear LP Shaft Cone Length	m	0.0568603
HP Shaft Cone Length	m	0.0574368
Rear HP Shaft Length	m	0.0823245
Rear HP Shaft Radius	m	0.0201499
Engine Length	m	1.45375
Max Engine Diameter	m	0.727549
Nacelle Length (Bypass only)	m	1.05737
LP Shaft Mass	kg	2.07345
HP Shaft Mass	kg	1.16514
Net Mass	kg	176.17
Total Mass	kg	229.021
LP Spool Inertia	kg*m²	0.997252
HP Spool Inertia	kg*m²	0.0434421

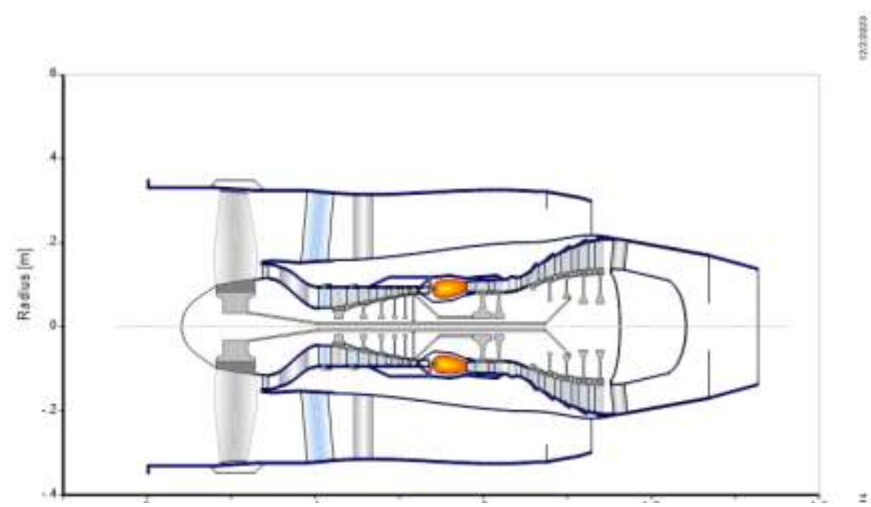

Figure 4: GasTurb 14 generated schematic for 2-spool engine architecture

Table 2: Geometric design parameters of GE9x engine

Design Parameter	Value (Units)	Justification
Engine Length	5.69 (m)	Aircraft Database GE9x
Engine Width	4.1 (m)	
Engine Height	4.16 (m)	
Fan Diameter	3.4 (m)	**GE9x-105B1A3 Engine Varient**
Engine Dry Weight	9630 (Kg)	
Fan Blade Count	16	
Fan Stage Count	1	
HPC Stage Count	11	
LPC Stage Count	3	
HPT Stage Count	2	
LPT Stage Count	6	

3-Spool Target Engine Specifications

Presented in Table *3* are the baseline engine performance parameters for the RR Trent XWB engine. Similar to the analysis in “2-Spool Target Engine Specifications”, some cycle efficiency parameters were deduced from the book “Propulsion and Power: An Exploration of Gas Turbine by Performance Modeling” by Joachim Kurzke and Ian Halliwell while other parameters such as component pressure ratios, air mass flow, and bypass ratio, and other cycle efficiencies were obtained from the case study conducted by Xue et al where they utilized the Trent XWB as their baseline engine model for design point computations and optimizations [1], [8]. The take-off net thrust and the inlet air mass flow rates were obtained from a online database linked in the table.

Table 3: RR Trent XWB engine baseline parameters

Design Parameter	Value (Units)	Justification
Net Thrust (FN) at **Take-Off**	97005 (lbf) / 431.5 (KN)	Aircraft Database Trent XWB-97
Inlet Air Mass Flow Rate (W)	1440 (Kg/s)	[8]
Overall Pressure Ratio (OPR)	50:1	Aircraft Database Trent XWB-97
IPC Pressure Ratio	8.4:1	[8]
HPC Pressure Ratio	4.5:1	[8]
Turbine Inlet Temperature (T4)	1882 (K)	[8]
Outer Fan Pressure Ratio (OFPR)	1.6	[8]
Inner Fan Pressure Ratio (IFPR)	1.4	[8]
Bypass Ratio (BPR)	9.3:1	[8]
Fan Inner/Outer Efficiencies (Polytropic)	0.91	[1]
HPC Efficiency (Polytropic)	0.90	[8]
LPC Efficiency (Polytropic)	0.92	[8]
IPC Efficiency (Polytropic)	0.90	[8]
LPT Efficiency (Polytropic)	0.8782	[8]
IPT Efficiency (Polytropic)	0.8901	[8]
HPT Efficiency (Polytropic)	0.89	[8]
Mechanical Efficiency	0.98	[2]

3-Spool Target Engine Architecture

The RR Trent XWB engine is made using a three-spool or three-shaft unmixed exhaust engine design architecture. The three-shaft concept, shown in Figure 5, unlike the two-shaft configuration used in the GE9x separates the fan from the IPC stages of the engine. This translates to a shaft called N1 connecting the fan by itself to the LPT stages, a shaft called N2 connecting the IPC and IPT stages together, and a final shaft called N3 connecting the HPC and HPT stages together. Sensibly, N1 will be the outer most shaft, N2 will be the intermediate shaft and N3 will be the inner most shaft. As briefly discussed in “2-Spool Target Engine Architecture” the three-shaft configuration optimizes performance by allowing improved control of the fan rotation speed, however, the implementation of this configuration is quite complex and costly and adds more mechanical risk and maintenance complexity therein. Depending on multiple factors such as maintenance time, cost, performance, safety among others, airlines must decide which configuration suits their protocols.

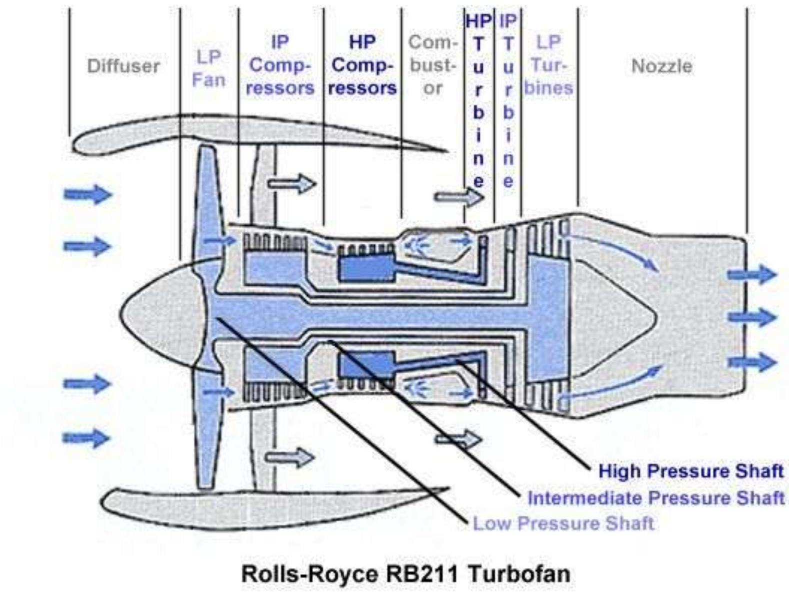

Figure 5: Simple schematic of a three-shaft turbofan engine configuration

To fully detail the engine architecture of the RR Trent XWB engine, a comprehensive schematic, driven from GasTurb 14, was provided in Figure 6. As discussed in "2-Spool Target Engine Architecture", the dimensions of the engine provided in Figure 6 will not be manipulated as the main influence on the thermodynamic cycle of the gas turbine is affected mainly by the engine performance parameters. The dimensions and geometric properties of the Trent XWB-97 are summarized in Table 4 for reference.

Front LP Shaft Cone Length	m	0.0128015
Middle LP Shaft Length	m	0.920848
Middle LP Shaft Radius	m	0.0183406
Front IP Shaft Cone Length	m	0.0532447
Middle IP Shaft Length	m	0.179406
Middle IP Shaft Radius	m	0.0283406
Rear LP Shaft Cone Length	m	0.0451147
Front HP Shaft Cone Length	m	0
Rear HP Shaft Cone Length	m	0.0737665
Rear HP Shaft Length	m	0.0666195
Rear HP Shaft Radius	m	0.0334342
Engine Length	m	2.18126
Max Engine Diameter	m	0.806492
Nacelle Length (Bypass only)	m	1.42678
LP Shaft Mass	kg	4.82092
IP Shaft Mass	kg	2.49071
HP Shaft Mass	kg	1.94253
Net Mass	kg	441.867
Total Mass	kg	574.427
LP Spool Inertia	kg*m²	5.80386
IP Spool Inertia	kg*m²	0.430426
HP Spool Inertia	kg*m²	0.0807676

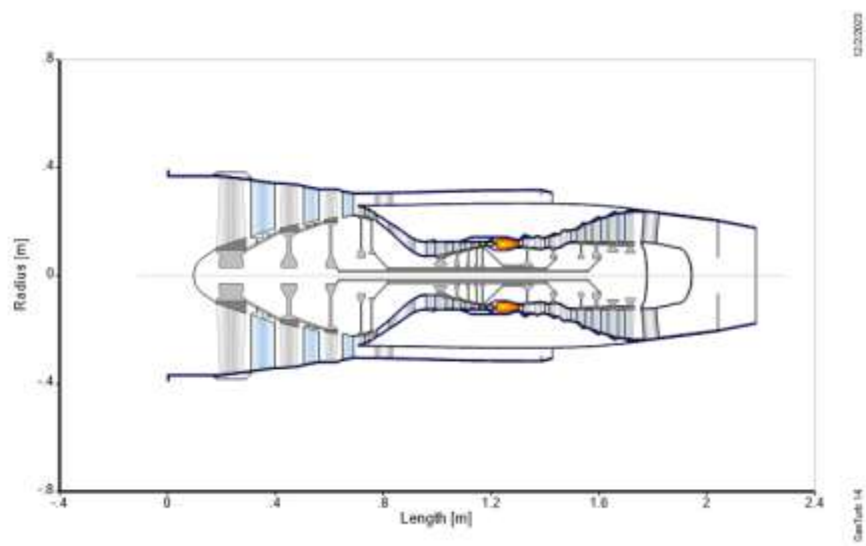

Figure 6: GasTurb 14 generated schematic for 3-spool engine architecture

Table 4: Geometric design parameters of RR Trent XWB engine

Design Parameter	Value (Units)	Justification
Engine Length	5.81 (m)	Aircraft Database Trent XWB-97 **Trent XWB-97 Engine Variant**
Engine Dry Weight	7549 (Kg)	
Fan Blade Count	18	
Fan Stage Count	1	
HPC Stage Count	6	
IPC Stage Count	8	
HPT Stage Count	1	
LPT Stage Count	6	
IPT Stage Count	2	

Engine Cycle Design Process through GasTurb 14

2-Spool Engine Design Process

Iteration # 1: Baseline Engine Design

Iteration Objective

The goal from this iteration attempt is to obtain the initial net thrust of the GE9x and the associated SFC, and then to run an iteration that targets the required net thrust of 150 KN while computing the corresponding inlet mass flow rate to fulfill that objective.

Design Process

After running GasTurb 14, the two-spool unmixed turbofan configuration is selected, then the “performance” tab is selected, and the associated data file is opened in GasTurb 14. After the design point window launches, performance parameters for the GE9x engine are integrated in place of the given data as seen in Figure 7. The altered parameters in accordance with Table 1 are boxed as shown. The inner and outer fan pressure ratios are assumed to be equal as was shown in [2].

Property	Unit	Value	Comment
Intake Pressure Ratio		0.99	
No (0) or Average (1) Core dP/P		1	
Inner Fan Pressure Ratio		1.5	
Outer Fan Pressure Ratio		1.5	
Compr. Interduct Press. Ratio		0.99	
HP Compressor Pressure Ratio		27	
Bypass Duct Pressure Ratio		0.98	
Turb. Interd. Ref. Press. Ratio		0.98	
Design Bypass Ratio		10	
Burner Exit Temperature	K	2200	
Burner Design Efficiency		0.9995	
Burner Partload Constant		1.6	used for off design only
Fuel Heating Value	MJ/kg	43.124	
Overboard Bleed	kg/s	0	
Power Offtake	kW	50	
HP Spool Mechanical Efficiency		0.99	
LP Spool Mechanical Efficiency		1	
Burner Pressure Ratio		0.97	
Turbine Exit Duct Press Ratio		0.98	

Figure 7: Basic data window where design parameters are modified in GasTurb 14

The next step is to define the ambient conditions. For this task, an airport must be selected to define the ambient conditions from parameters such as the altitude of the airport above sea level, the relative humidity, and the change in temperature from ISA conditions. The chosen airport for this task is Washington Dulles International Airport (IAD). Through a quick online search, it can be noted that the relative humidity in IAD is 66%, the altitude above sea level is 95.2 meters and the change in temperature from ISA is around 0.5 Kelvin. Since the engine is modeled at takeoff conditions, the local Mach number is kept at 0. The modifications mentioned are displayed in Figure 8.

Property	Unit	Value	Comment
Altitude	m	95.2	
Delta T from ISA	K	0.5	
Relative Humidity [%]		66	
Mach Number		0	

Figure 8: Ambient conditions at IAD for takeoff conditions

The next step is to define the desired inlet mass flow rate and the component polytropic efficiencies for the GE9x baseline engine cycle in accordance with Table 1. The modifications are displayed in Figure 9.

Sensitivity to | a selected input property

Property	Unit	Value	Comment
Inlet Corr. Flow W2Rstd	kg/s	1500	

Property	Unit	Value	Comment
Polytr.Inner LPC Efficiency		0.91	
Polytr.Outer LPC Efficiency		0.91	

Sensitivity to | a selected input property

Property	Unit	Value	Comment
Polytr.HPC Efficiency		0.91	

Property	Unit	Value	Comment
Polytr. HPT Efficiency		0.89	

Property	Unit	Value	Comment
Polytr. LPT Efficiency		0.91	

Figure 9: Component efficiency and mass flow rate modifications for baseline engine cycle

Now that the relevant cycle parameters, component efficiencies, mass flow rate, and ambient conditions are defined, the cycle design point calculation is performed by clicking the “Design Point” tab. The resulting summary is displayed in Figure 10. The TSFC and the resulting net thrust from the cycle are boxed for later analysis. The mass flow is also boxed for later reference upon completion of the iteration function.

```
             W          T        P         WRstd
Station   kg/s          K       kPa        kg/s        FN        =     464.51 kN
 amb                 288.03   100.187
  1     1465.511     288.03   100.187                  TSFC      =    10.7621 g/(kN*s)
  2     1465.511     288.03    99.185   1500.000       WF        =    4.99905 kg/s
 13     1332.283     326.98   148.777    968.614       BPR       =    10.0000
 21      133.228     326.98   148.777     96.861       s NOx     =     1.6623
 25      133.228     326.98   147.289     97.840       Core Eff  =     0.5044
  3      129.231     881.79  3976.811      5.772       Prop Eff  =     0.0000
 31      113.244     881.79  3976.811                  P3/P2     =      40.10
  4      118.243    2200.00  3857.507      8.599       P2/P1     =     0.9900
405      123.572    2150.00  3857.507                  NGV Out.  2 Stage HPT
 41      126.237    2126.45  3857.507      9.026       P16/P13   =     0.9800
 43      126.237    1667.87  1070.851                  P25/P21   =     0.9900
 44      132.898    1632.66  1070.851                  P45/P44   =     0.9800
 45      132.898    1632.66  1049.434     30.604       P6/P5     =     0.9800
 49      132.898    1298.61   336.976                  A8        =    0.38552 m²
  5      136.895    1282.05   336.976     86.999       A18       =    4.53831 m²
  8      136.895    1282.05   330.237     88.774       P8/Pamb   =    3.29622
 18     1332.283     326.98   145.802    988.383       P18/Pamb  =    1.45530
Bleed      1.332     881.79  3976.803                  WBld/W25  =    0.01000
-------------------------------------------------      CD8       =    0.98000
Efficiencies:     isentr  polytr    RNI     P/P        CD18      =    0.95820
 Outer LPC        0.9047  0.9100  0.979   1.500        XM8       =    1.00000
 Inner LPC        0.9047  0.9100  0.979   1.500        XM18      =    0.75250
 HP Compressor    0.8652  0.9100  1.251  27.000        V18/V8,id=     0.29505
 Burner           0.9995                  0.970        Loading   =     100.00 %
 HP Turbine       0.9030  0.8900  3.661   3.602        e444 th   =    0.88707
 LP Turbine       0.9200  0.9100  1.361   3.114        PWX       =      50.00 kW
-------------------------------------------------      WCHN/W25  =    0.06000
HP Spool mech Eff 0.9900   Nom Spd    7507 rpm         WCHR/W25  =    0.05000
LP Spool mech Eff 1.0000   Nom Spd   13500 rpm         WLcl/W25  =    0.03000
-------------------------------------------------
hum [%]      war0       FHV      Fuel
  66.0    0.00704     43.124    Generic
```

Figure 10: Summary of baseline cycle design point calculation for GE9X engine

After the net thrust and the SFC of the cycle have been obtained, an iteration should be performed to obtain the required mass flow rate that would yield the target net thrust of 150 KN. To perform an iteration, the summary window is first closed, and the "Iterations" tab is selected from the "additional" drop down menu. Once the iterations window is opened, the input variable defined is set to be the inlet mass flow rate, and the target value is the net thrust which is set to a value of 150 KN as expected. Once the variables and values are set, the box should be checked to activate the iteration. The iteration setup is displayed in Figure 11.

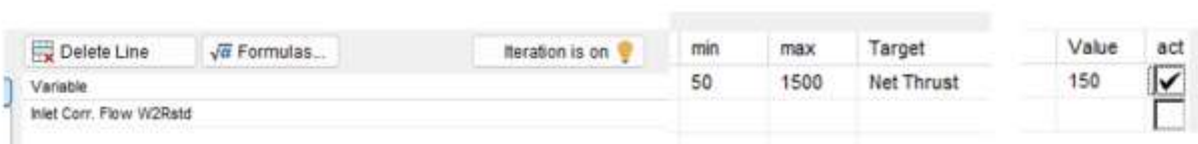

Figure 11: Iteration setup

The iteration window is then closed and the cycle point calculation, with the iteration activated, is computed again and the summary of the computation is displayed in Figure 12. The new mass flow rate and the target net thrust are boxed for later analysis.

```
             W           T          P          WRstd
Station    kg/s          K         kPa         kg/s           FN          =     150.00 kN
 amb                   288.03    100.187
  1      473.373       288.03    100.187                      TSFC        =    10.7649 g/(kN
  2      473.373       288.03     99.185     484.513          WF          =    1.61474 kg/s
 13      430.339       326.98    148.777     312.871          BPR         =    10.0000
 21       43.034       326.98    148.777      31.287          s NOx       =     1.6623
 25       43.034       326.98    147.289      31.603          Core Eff    =     0.5038
  3       41.743       881.79   3976.811       1.864          Prop Eff    =     0.0000
 31       36.579       881.79   3976.811                      P3/P2       =      40.10
  4       38.194      2200.00   3857.507       2.778          P2/P1       =     0.9900
405       39.915      2150.00   3857.507                      NGV Out.  2 Stage HPT
 41       40.776      2126.45   3857.507       2.915          P16/P13     =     0.9800
 43       40.776      1667.24   1068.754                      P25/P21     =     0.9900
 44       42.927      1632.06   1068.754                      P45/P44     =     0.9800
 45       42.927      1632.06   1047.379       9.903          P6/P5       =     0.9800
 49       42.927      1297.98    336.155                      A8          =    0.12480 m²
  5       44.218      1281.44    336.155      28.163          A18         =    1.46591 m²
  8       44.218      1281.44    329.432      28.738          P8/Pamb     =    3.28819
 18      430.339       326.98    145.802     319.256          P18/Pamb    =    1.45530
Bleed      0.430       881.79   3976.803                      WBld/W25    =    0.01000
---------------------------------------------                 CD8         =    0.98000
Efficiencies:     isentr   polytr    RNI     P/P              CD18        =    0.95820
 Outer LPC        0.9047   0.9100   0.979   1.500             XM8         =    1.00000
 Inner LPC        0.9047   0.9100   0.979   1.500             XM18        =    0.75250
 HP Compressor    0.8652   0.9100   1.251  27.000             V18/V8,id=       0.29538
 Burner           0.9995                    0.970             Loading     =     100.00 %
 HP Turbine       0.9030   0.8900   3.661   3.609             e444 th     =    0.88709
 LP Turbine       0.9200   0.9100   1.359   3.116             PWX         =      50.00 kW
---------------------------------------------                 WCHN/W25    =    0.06000
HP Spool mech Eff 0.9900   Nom Spd   13208 rpm                WCHR/W25    =    0.05000
LP Spool mech Eff 1.0000   Nom Spd   13500 rpm                WLcl/W25    =    0.03000
---------------------------------------------
hum [%]      war0        FHV       Fuel
  66.0     0.00704     43.124     Generic
```

Figure 12: Summary of cycle design point with iteration an activated iteration

As seen from Figure 12, the resulting mass flow rate is 484.513 Kg/s. This value will be the new mass flow rate input used to perform the simulations in

Observations and Discussion

The cycle design point performed the first time yielded a net thrust of 464.51 KN and a TSFC of 10.7621 g/(KN*s). This result seems plausible given that the net thrust for the GE9x at takeoff is 489.3 KN. The discrepancy is caused from uncertainties and assumptions in defining the cycle parameters and efficiencies as well as the absence of a complete off-design analysis which would give valuable insight on the engine's operating performance through different flight stages and so on. The SFC value was also in a reasonable range compared to the target SFC which begs the investigation of the calculations done and assumptions made by the authors in [2] to suggest a value as such. Overall, the results are accepted factoring in the previous assumptions and estimations.

Iteration # 2: SFC Analysis through a Parametric Study

Iteration Objective

The goal of this design iteration is to perform a parametric study where the optimum outer fan pressure ratio is found to achieve the lowest possible SFC rating and the highest net thrust while also computing the ideal jet velocity ratio that will yield the best engine performance in terms of thrust and SFC. The iteration function used in Iteration # 1 is still expected to be running to keep the net thrust at 150 KN by changing the inlet mass flow rate when the OFFPR is changed to its optimum value.

Design Process

To conduct a parametric study, the summarized result of the cycle generated in Iteration # 1 should be closed and the tab "Parametric" should be selected from the "Task" drop down menu. Once the Parametric study window is opened, the design bypass ratio was selected as the varying parameter. The setup is shown in Figure 13.

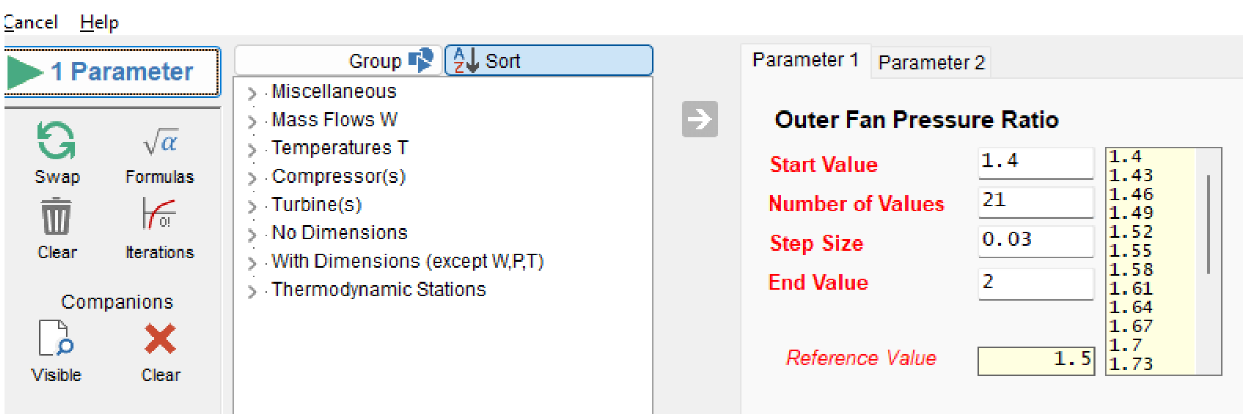

Figure 13: Parametric study initial setup window

After running the parametric study, the outer fan pressure ratio was set as the independent variable and the SFC, the net thrust and the ideal jet velocity ratio were set as the dependent variables as seen in Figure 14. After the axes have been defined, the parametric study was performed through clicking the "Draw y=f(x)" tab. The resulting graph is displayed in Figure 15.

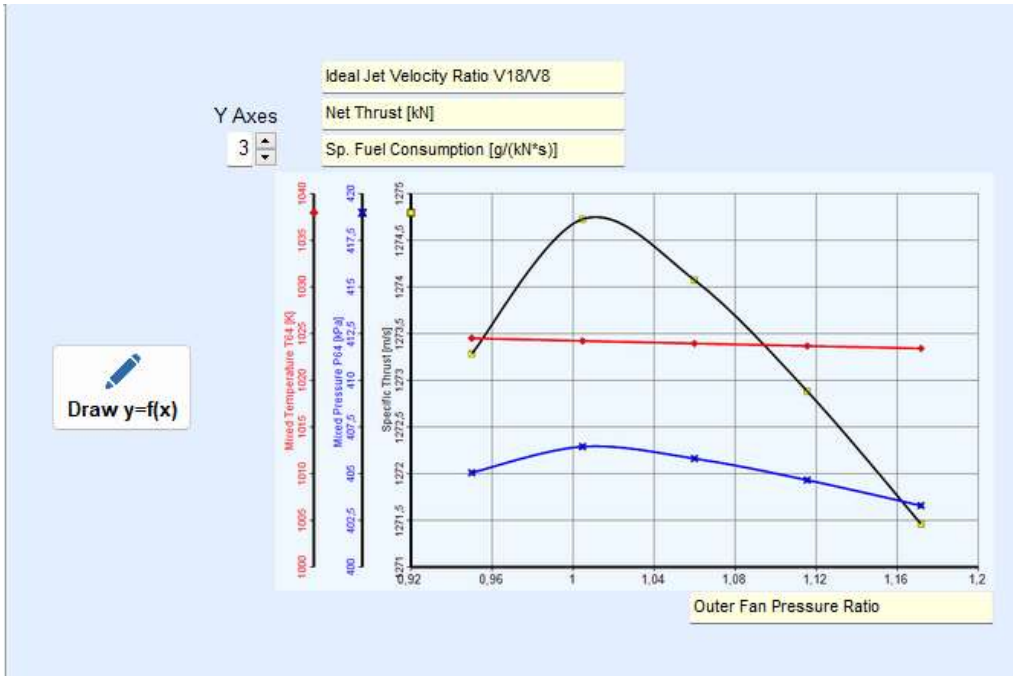

Figure 14: Axis labeling for parametric study

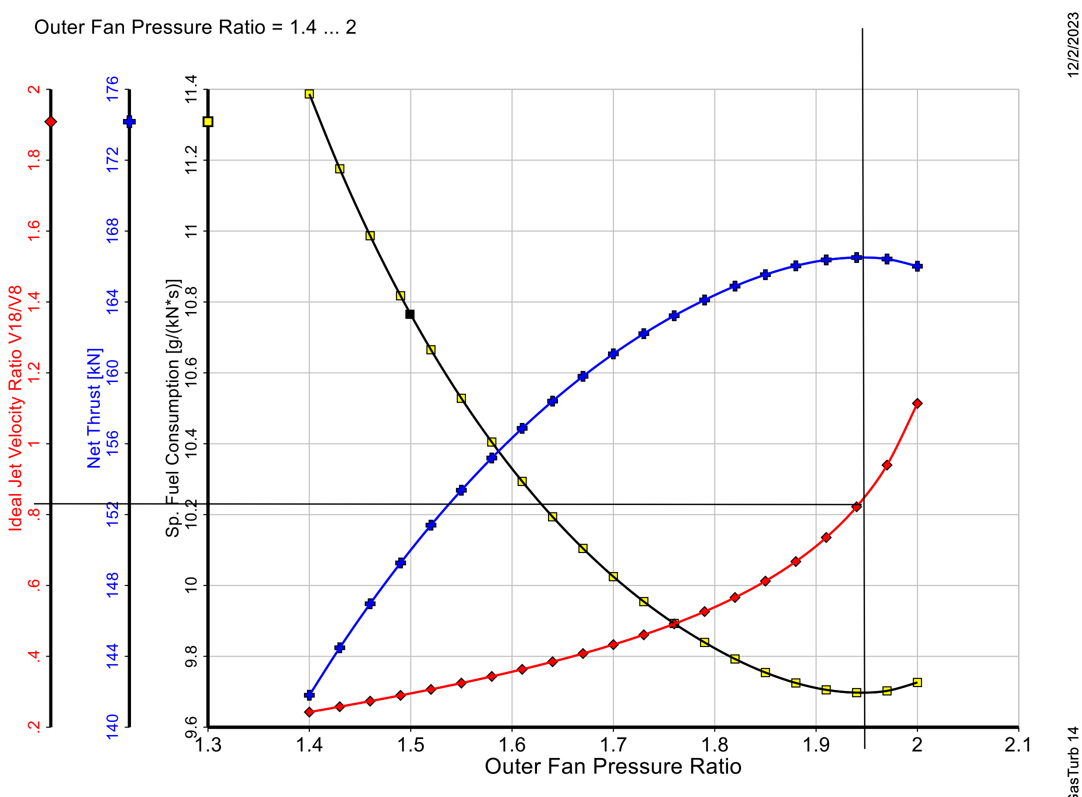

Figure 15: Parametric study showing the optimum OFPR and jet velocity ratio for achieving lowest SFC and highest net thrust

As seen from Figure 15, the optimum value for the OFPR is around 1.95 to achieve the highest possible net thrust and the lowest possible SFC for the engine cycle. The corresponding jet velocity ratio for such behavior amounts to around 0.85. The next step in this iteration is to find the resulting SFC by conducting a design point after changing the value for the OFPR while also making sure that the iteration function done in iteration # 1 is enabled. After changing the OFPR to 1.95 and running the design point and reenabling the iteration function, the design point simulation was conducted, and the summary of the corresponding results Is shown in Figure 16. The new mass flow rate, ideal jet velocity ratio, SFC and net thrust are all boxed for later analysis.

Station	W kg/s	T K	P kPa	WRstd kg/s
amb		288.03	100.187	
1	426.499	288.03	100.187	
2	426.499	288.03	99.185	436.536
13	387.726	354.91	193.410	225.907
21	38.773	326.98	148.777	28.189
25	38.773	326.98	147.289	28.474
3	37.609	881.79	3976.811	1.680
31	32.957	881.79	3976.811	
4	34.412	2200.00	3857.507	2.503
405	35.962	2150.00	3857.507	
41	36.738	2126.45	3857.507	2.627
43	36.738	1667.14	1068.414	
44	38.677	1631.96	1068.414	
45	38.677	1631.96	1047.046	8.925
49	38.677	1071.81	134.427	
5	39.840	1061.08	134.427	57.740
8	39.840	1061.08	131.739	58.918
18	387.726	354.91	189.542	230.518
Bleed	0.388	881.79	3976.803	

Efficiencies:	isentr	polytr	RNI	P/P
Outer LPC	0.9012	0.9100	0.979	1.950
Inner LPC	0.9047	0.9100	0.979	1.500
HP Compressor	0.8652	0.9100	1.251	27.000
Burner	0.9995			0.970
HP Turbine	0.9030	0.8900	3.661	3.610
LP Turbine	0.9279	0.9100	1.358	7.789

HP Spool mech Eff 0.9900 Nom Spd 13915 rpm
LP Spool mech Eff 1.0000 Nom Spd 13500 rpm

hum [%]	war0	FHV	Fuel
66.0	0.00704	43.124	Generic

FN	=	150.00 kN
TSFC	=	9.6990 g/(kN*s)
WF	=	1.45484 kg/s
BPR	=	10.0000
s NOx	=	1.6623
Core Eff	=	0.5037
Prop Eff	=	0.0000
P3/P2	=	40.10
P2/P1	=	0.9900
NGV Out.	2	Stage HPT
P16/P13	=	0.9800
P25/P21	=	0.9900
P45/P44	=	0.9800
P6/P5	=	0.9800
A8	=	0.29598 m²
A18	=	0.98023 m²
P8/Pamb	=	1.31493
P18/Pamb	=	1.89189
WBld/W25	=	0.01000
CD8	=	0.95653
CD18	=	0.97530
XM8	=	0.65667
XM18	=	0.99984
V18/V8,id	=	0.85782
Loading	=	100.00 %
e444 th	=	0.88710
PWX	=	50.00 kW
WCHN/W25	=	0.06000
WCHR/W25	=	0.05000
WLcl/W25	=	0.03000

Figure 16: Summary of cycle computation after parametric study

Observations and Discussion

As seen from Figure 16, the new SFC value after the parametric study was conducted in the second iteration is 9.6990 g/(KN*s) which is a 10% drop from the SFC value found in the first iteration (Figure 12) which was 10.7649 g/(KN*s). The second observation was that the value of the ideal jet velocity ratio optically measured in Figure 15 was validated in Figure 16 when the OFPR is set to 1.95. In summary the results agree with established theory and are therefore acceptable.

Iteration # 3: SFC Analysis through Cycle Optimization

Iteration Objective

The goal of this iteration attempt is for the optimization of the engine cycle for the main purpose of lowering the SFC as much as possible. The design parameters and the constraints along with the constraints are influenced by the work of Fanzago et al where they conducted a design optimization using GasTurb 13 on a two-spool turbofan engine [2].

Design Process

First, the design point computation from iteration # 2 is closed, then, from the "Task" drop down menu, the "optimize" tab is selected and launched. Once the optimization window is opened, the design variables of interest along with the design constraints are listed, and the figure of merit is selected to minimize SFC. This process is illustrated in Figure 17 and the optimization results are shown in Figure 18.

Variables	Min Value	Start Value	Max Value
Inner Fan Pressure Ratio	1.1	1.5	1.975
Outer Fan Pressure Ratio	1.1	1.95	1.975
Design Bypass Ratio	9	10	12
HP Compressor Pressure Ratio	20	27	32
Burner Exit Temperature	1800	2200	2500

Constraints	Min Value	Start Value	Max Value
Overall Pressure Ratio P3/P2	30	40.095	45
LPT Inlet Temperature T45	1400	1631.96	1800
HPT Pressure Ratio	3	3.6105	4.5
LPT Pressure Ratio	7	7.78895	8
HPC Exit Temperature T3	800	881.791	900

Figure of Merit

Maximize Minimize

Sp. Fuel Consumption

Figure 17: Engin cycle optimization setup

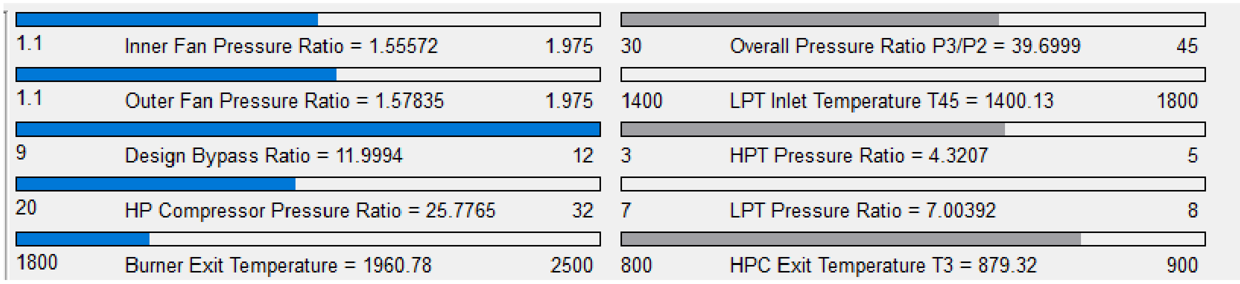

Figure 18: Optimization results after several iterations

After the optimization was completed, the cycle was run again from the design point and the summarized results are displayed in .

```
              W          T         P        WRstd
Station   kg/s         K        kPa       kg/s        FN         =     150.00 kN
 amb                 288.03   100.187
  1     515.606    288.03   100.187                   TSFC       =     8.2485 g/(kN*s)
  2     515.606    288.03    99.185    527.740        WF         =    1.23727 kg/s
 13     473.947    335.24   161.143    322.125        BPR        =    11.3769
 21      41.659    324.92   145.791     30.810        s NOx      =     1.8769
 25      41.659    324.92   144.333     31.121        Core Eff   =     0.5019
  3      40.409    899.84  4271.614      1.697        Prop Eff   =     0.0000
 31      35.410    899.84  4271.614                   P3/P2      =      43.07
  4      36.647   1982.89  4143.466      2.356        P2/P1      =     0.9900
405      38.314   1940.55  4143.466                   NGV Out.  2 Stage HPT
 41      39.147   1920.64  4143.466      2.476        P16/P13    =     0.9800
 43      39.147   1424.60   900.513                   P25/P21    =     0.9900
 44      41.230   1400.39   900.513                   P45/P44    =     0.9800
 45      41.230   1400.39   882.503     10.457        P6/P5      =     0.9800
 49      41.230    924.60   125.598                   A8         =    0.34432 m²
  5      42.480    917.88   125.598     61.287        A18        =    1.44856 m²
  8      42.480    917.88   123.086     62.538        P8/Pamb    =    1.22857
 18     473.947    335.24   157.920    328.699        P18/Pamb   =    1.57626
Bleed     0.417    899.84  4271.614                   WBld/W25   =    0.01000
---------------------------------------------         CD8        =    0.95024
Efficiencies:    isentr  polytr    RNI    P/P         CD18       =    0.96523
 Outer LPC       0.9037  0.9100  0.979  1.625         XM8        =    0.56414
 Inner LPC       0.9050  0.9100  0.979  1.470         XM18       =    0.83352
 HP Compressor   0.8639  0.9100  1.235 29.595         V18/V8,id=      0.88137
 Burner          0.9995                 0.970         Loading    =     100.00 %
 HP Turbine      0.9058  0.8900  4.452  4.601         e444 th    =    0.88650
 LP Turbine      0.9277  0.9100  1.369  7.026         PWX        =      50.00 kW
---------------------------------------------         WCHN/W25   =    0.06000
HP Spool mech Eff 0.9900  Nom Spd  13310 rpm          WCHR/W25   =    0.05000
LP Spool mech Eff 1.0000  Nom Spd  13500 rpm          WLcl/W25   =    0.03000
---------------------------------------------
hum [%]      war0       FHV      Fuel
  66.0    0.00704    43.124    Generic
```

Figure 19: Summary of cycle design point after optimization

Observations and Discussion

As seen from Figure 19, the SFC is 8.2485 g/ (KN*s) after the optimization was completed. The SFC has dropped by 15% compared to the SFC from iteration # 2 which was 9.6990 g/(KN*s). This shows a promising merit in optimization utilization and with improved experience and understanding of the thermodynamic cycle and the gas turbine cycle, better optimizations can be performed and lower SFC can be expected. Since the optimization process was indeed successful and the results met expectations, they were acceptable.

3-Spool Engine Design Process

Iteration # 1: Baseline Engine Design

The design process and objectives are identical to what was done in the two-spool configurations so most of the content in this part of the report will directly showcase the results. Papers [1], and p8[were extensively useful in performing the iterations required for the three-spool engine configuration.

Design Process Results

As stated, the step-by-step design process is represented in the “2-Spool Engine Design Process” section of this report. The final results relevant for the analysis in the “Observation and Discussion" sections of the remainder of this report are directly displayed in the subsequent section titled “ Design Process Results” for all the proceeding iterations.

Property	*Unit*	*Value*	*Comment*
Intake Pressure Ratio		0.99	
No (0) or Average (1) Core dP/P		1	
Inner Fan Pressure Ratio		1.4	
Outer Fan Pressure Ratio		1.6	
Core Inlet Duct Press. Ratio		1	
IP Compressor Pressure Ratio		2	
Compr. Interduct Press. Ratio		0.98	
HP Compressor Pressure Ratio		4.5	
Bypass Duct Pressure Ratio		0.975	
Inlet Corr. Flow W2Rstd	kg/s	1440	
Design Bypass Ratio		9.3	
Burner Exit Temperature	K	1882	
Burner Design Efficiency		0.9995	
Burner Partload Constant		1.6	used for off design only
Fuel Heating Value	MJ/kg	43.124	
Overboard Bleed	kg/s	0	
Power Offtake	kW	50	
HP Spool Mechanical Efficiency		0.98	
IP Spool Mechanical Efficiency		1	
LP Spool Mechanical Efficiency		1	
Burner Pressure Ratio		0.95	
IPT Interd. Ref. Press. Ratio		0.992	
LPT Interd. Ref. Press. Ratio		1	
Turbine Exit Duct Press Ratio		0.99	

Figure 20: Basic Data for three-spool configuration

Property	*Unit*	*Value*	*Comment*
Altitude	m	95.2	
Delta T from ISA	K	0.5	
Relative Humidity [%]		66	
Mach Number		0	

Figure 21: Ambient conditions

Property	*Unit*	*Value*	*Comment*
Altitude	m	95.2	
Delta T from ISA	K	0.5	
Relative Humidity [%]		66	
Mach Number		0	

Property	*Unit*	*Value*	*Comment*
Polytr.Inner LPC Efficiency		0.92	
Polytr.Outer LPC Efficiency		0.92	

Property	*Unit*	*Value*	*Comment*
Polytr.IPC Efficiency		0.9	

Property	*Unit*	*Value*	*Comment*
Polytr.HPC Efficiency		0.9	

Property	*Unit*	*Value*	*Comment*
Polytr. HPT Efficiency		0.89	

Property	*Unit*	*Value*	*Comment*
Polytr.IP Turbine Efficiency		0.8901	

Property	*Unit*	*Value*	*Comment*
Polytr. LPT Efficiency		0.8782	

Figure 22: Efficiency ratings and ambient conditions for 3-spool configuration

```
             W          T         P        WRstd       FN        =    415.76 kN
Station   kg/s          K        kPa       kg/s       TSFC      =   10.8940 g/(kN*s)
 amb                 288.03    100.187                 WF        =   4.52928 kg/s
  2    1406.891      288.03     99.185   1440.000     s NOx     =   0.28785
 13    1270.299      333.11    158.696    873.901     BPR       =    9.3000
 21     136.591      319.65    138.859    105.199     Core Eff  =    0.3967
 22     136.591      319.65    138.859    105.199     Prop Eff  =    0.0000
 24     136.591      397.87    277.717     58.684     P3/P2     =    12.348
 25     136.591      397.87    272.163     59.881     P2/P1     =   0.99000
  3     131.811      632.51   1224.733     16.191     P22/P21   =   1.00000
 31     116.786      632.51   1224.733                P25/P24   =   0.98000
  4     121.315     1882.00   1163.496     27.054     P4/P3     =   0.95000
 41     128.144     1823.49   1163.496     28.129     P44/P43   =   0.99200
 42     128.144     1625.61    642.218                P48/P47   =   1.00000
 43     136.340     1572.66    642.218                P6/P5     =   0.99000
 44     136.340     1572.66    637.080                P16/P13   =   0.97500
 45     139.072     1554.76    637.080     51.482     P16/P6    =   1.16114
 46     139.072     1494.83    521.982                P5/P2     =   1.35708
 47     139.755     1484.09    521.982                V18/V8,id=   0.66200
 48     139.755     1484.09    521.982     61.690     A8        =   1.06466 m²
 49     139.755     1131.58    134.602                A18       =   4.02239 m²
  5     141.121     1126.80    134.602    210.495     XM8       =   0.67106
  8     141.121     1126.80    133.256    212.621     XM18      =   0.81339
 18    1270.299      333.11    154.728    896.310     WBld/W2   =   0.00000
Bleed     0.000      632.51   1224.736
-----------------------------------------------
Efficiencies:    isentr  polytr    RNI     P/P      CD8       =   0.94907
 Outer LPC       0.9145  0.9200  0.979   1.600     CD18      =   0.95443
 Inner LPC       0.9161  0.9200  0.979   1.400     PWX       =     50.00 kW
 IP Compressor   0.8899  0.9000  1.212   2.000     WlkLP/W25=   0.00000
 HP Compressor   0.8780  0.9000  1.829   4.500     WBld/W25 =   0.00000
 Burner          0.9995                  0.950     Loading   =    100.00 %
 HP Turbine      0.8962  0.8900  1.327   1.812     e442 th   =   0.87810
 IP Turbine      0.8922  0.8901  0.876   1.221     WCHN/W25 =   0.05000
 LP Turbine      0.8942  0.8782  0.757   3.878     WCHR/W25 =   0.06000
-----------------------------------------------     WCIN/W25 =   0.02000
HP Spool mech Eff 0.9800  Nom Spd   8574 rpm       WCIR/W25 =   0.00500
IP Spool mech Eff 1.0000  Nom Spd  15000 rpm       WCLR/W25 =   0.01000
LP Spool mech Eff 1.0000  Nom Spd  12800 rpm
-----------------------------------------------
hum [%]     war0       FHV      Fuel
  66.0    0.00704    43.124    Generic
```

Figure 23: Summary of cycle design point for three-spool configuration

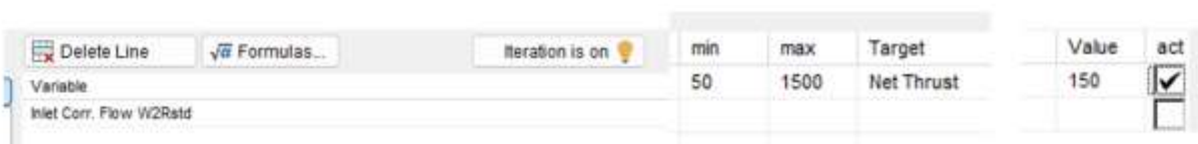

Figure 24: Iteration setup for three-spool configuration

```
                 W           T          P       WRstd
Station   kg/s           K        kPa       kg/s          FN         =     150.00 kN
 amb                 288.03    100.187                    TSFC       =    10.9004 g/(kN*s)
  2     507.886      288.03     99.185     519.839        WF         =    1.63507 kg/s
 13     458.577      333.11    158.696     315.478        s NOx      =    0.28785
 21      49.309      319.65    138.859      37.977        BPR        =     9.3000
 22      49.309      319.65    138.859      37.977        Core Eff   =     0.3962
 24      49.309      397.87    277.717      21.185        Prop Eff   =     0.0000
 25      49.309      397.87    272.163      21.617        P3/P2      =     12.348
  3      47.584      632.51   1224.733       5.845        P2/P1      =    0.99000
 31      42.159      632.51   1224.733                    P22/P21    =    1.00000
  4      43.795     1882.00   1163.496       9.766        P25/P24    =    0.98000
 41      46.260     1823.49   1163.496      10.155        P4/P3      =    0.95000
 42      46.260     1625.08    641.133                    P44/P43    =    0.99200
 43      49.219     1572.15    641.133                    P48/P47    =    1.00000
 44      49.219     1572.15    636.004                    P6/P5      =    0.99000
 45      50.205     1554.26    636.004      18.613        P16/P13    =    0.97500
 46      50.205     1494.33    521.067                    P16/P6     =    1.16380
 47      50.451     1483.59    521.067                    P5/P2      =    1.35398
 48      50.451     1483.59    521.067      22.306        V18/V8,id=    0.66474
 49      50.451     1131.06    134.294                    A8         =    0.38606 m²
  5      50.944     1126.29    134.294      76.145        A18        =    1.45208 m²
  8      50.944     1126.29    132.951      76.914        XM8        =    0.66827
 18     458.577      333.11    154.728     323.567        XM18       =    0.81339
Bleed     0.000      632.51   1224.736                    WBld/W2    =    0.00000
--------------------------------------------
Efficiencies:    isentr   polytr    RNI     P/P           CD8        =    0.94883
 Outer LPC       0.9145   0.9200   0.979   1.600          CD18       =    0.95443
 Inner LPC       0.9161   0.9200   0.979   1.400          PWX        =      50.00 kW
 IP Compressor   0.8899   0.9000   1.212   2.000          WlkLP/W25=      0.00000
 HP Compressor   0.8780   0.9000   1.829   4.500          WBld/W25 =      0.00000
 Burner          0.9995                    0.950          Loading    =     100.00 %
 HP Turbine      0.8963   0.8900   1.327   1.815          e442 th    =    0.87812
 IP Turbine      0.8923   0.8901   0.874   1.221          WCHN/W25 =      0.05000
 LP Turbine      0.8942   0.8782   0.756   3.880          WCHR/W25 =      0.06000
--------------------------------------------              WCIN/W25 =      0.02000
HP Spool mech Eff 0.9800   Nom Spd   14271 rpm            WCIR/W25 =      0.00500
IP Spool mech Eff 1.0000   Nom Spd   15000 rpm            WCLR/W25 =      0.01000
LP Spool mech Eff 1.0000   Nom Spd   12800 rpm
--------------------------------------------
hum [%]      war0        FHV       Fuel
  66.0     0.00704     43.124     Generic
```

Figure 25: Design point summary after iteration for three-spool configuration

Observations and Discussion

Similar to the results in the 2-spool setup for iteration # 1, the net thrust and the specific fuel consumption values agree reasonably well with acquired data and with the theory of gas turbine design, so this iteration is passable.

Iteration # 2: SFC Analysis through a Parametric Study

Design Process Results

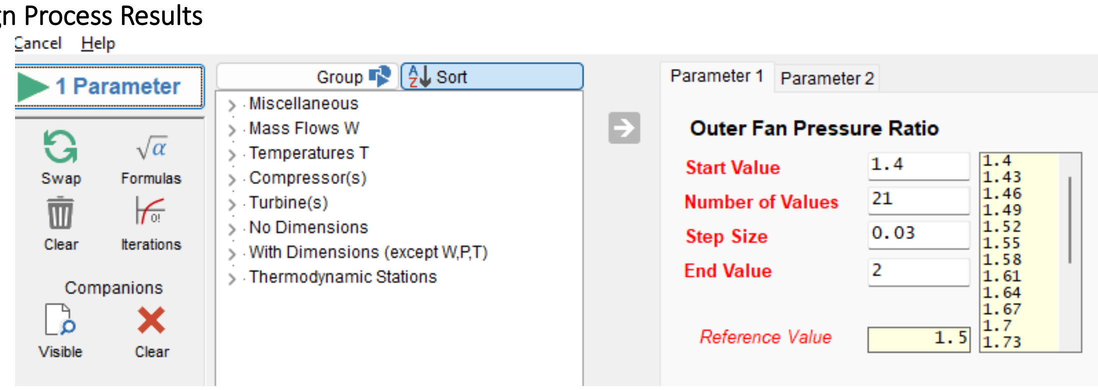

Figure 26: Parametric study for three-spool configuration

Figure 27: Parametric study setup for three-spool configuration (No Net thrust from list)

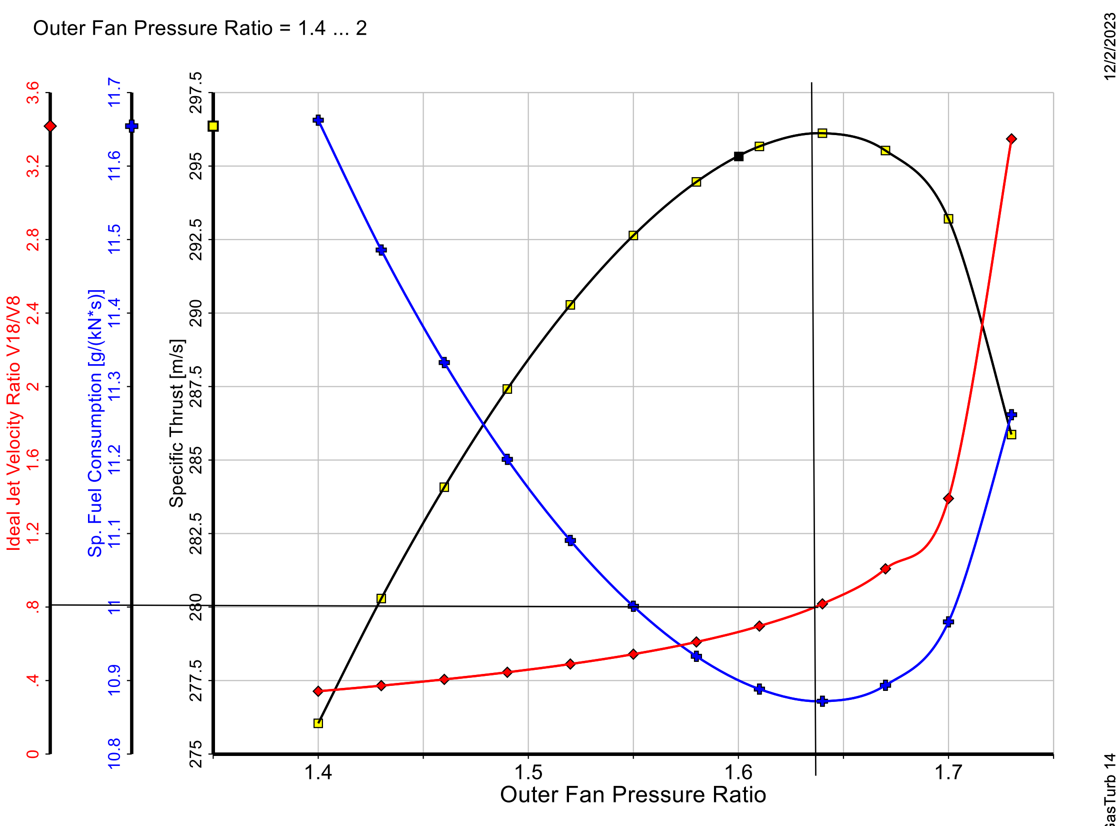

Figure 28: Parametric study results for three-spool configuration

The OFPR from Figure 28 is approximately 1.635 while the ideal jet velocity is around 0.8.

```
             W          T          P        WRstd
Station    kg/s         K         kPa       kg/s        FN        =     150.00 kN
 amb                 288.03    100.187                  TSFC      =    10.8723 g/(kN*s)
  2      506.577     288.03     99.185    518.499       WF        =    1.63085 kg/s
 13      457.395     335.34    162.167    308.959       s NOx     =    0.28785
 21       49.182     319.65    138.859     37.879       BPR       =     9.3000
 22       49.182     319.65    138.859     37.879       Core Eff  =     0.3962
 24       49.182     397.87    277.717     21.130       Prop Eff  =     0.0000
 25       49.182     397.87    272.163     21.561       P3/P2     =     12.348
  3       47.461     632.51   1224.733      5.830       P2/P1     =    0.99000
 31       42.051     632.51   1224.733                  P22/P21   =    1.00000
  4       43.682    1882.00   1163.496      9.741       P25/P24   =    0.98000
 41       46.141    1823.49   1163.496     10.128       P4/P3     =    0.95000
 42       46.141    1625.08    641.129                  P44/P43   =    0.99200
 43       49.092    1572.15    641.129                  P48/P47   =    1.00000
 44       49.092    1572.15    636.000                  P6/P5     =    0.99000
 45       50.075    1554.26    636.000     18.565       P16/P13   =    0.97500
 46       50.075    1494.33    521.063                  P16/P6    =    1.27871
 47       50.321    1483.59    521.063                  P5/P2     =    1.25926
 48       50.321    1483.59    521.063     22.248       V18/V8,id=    0.79377
 49       50.321    1114.44    124.900                  A8        =    0.45275 m²
  5       50.813    1109.82    124.900     81.062       A18       =    1.40845 m²
  8       50.813    1109.82    123.651     81.881       XM8       =    0.57387
 18      457.395     335.35    158.113    316.882       XM18      =    0.83471
Bleed      0.000     632.51   1224.736                  WBld/W2   =    0.00000
--------------------------------------------------
Efficiencies:    isentr   polytr    RNI     P/P        CD8       =    0.94081
 Outer LPC       0.9143   0.9200   0.979   1.635       CD18      =    0.95666
 Inner LPC       0.9161   0.9200   0.979   1.400       PWX       =      50.00 kW
 IP Compressor   0.8899   0.9000   1.212   2.000       WlkLP/W25=    0.00000
 HP Compressor   0.8780   0.9000   1.829   4.500       WBld/W25 =    0.00000
 Burner          0.9995                    0.950       Loading   =     100.00 %
 HP Turbine      0.8963   0.8900   1.327   1.815       e442 th   =    0.87812
 IP Turbine      0.8923   0.8901   0.874   1.221       WCHN/W25 =    0.05000
 LP Turbine      0.8951   0.8782   0.756   4.172       WCHR/W25 =    0.06000
--------------------------------------------------    WCIN/W25 =    0.02000
HP Spool mech Eff 0.9800   Nom Spd   14289 rpm        WCIR/W25 =    0.00500
IP Spool mech Eff 1.0000   Nom Spd   15000 rpm        WCLR/W25 =    0.01000
LP Spool mech Eff 1.0000   Nom Spd   12800 rpm
--------------------------------------------------
hum [%]      war0        FHV      Fuel
  66.0     0.00704     43.124    Generic
```

Figure 29: Summary of cycle design point after parametric study for 3-spool configuration

Observations and Discussion

As seen from the parametric study, a slight change in the OFPR from 1.6 to 1.635 decreased the SFC as seen in Figure 29 above. The SFC went down from 10.9004 g/(KN*s) to 10.8723 g/(KN*s). Therefore, the results were consistent and passable. Additionally, the ideal jet velocity was indeed around 0.8 as validated from the cycle calculation in Figure 29.

Iteration # 3: SFC Analysis through Cycle Optimization

Design Process Results

Variables	Min Value	Start Value	Max Value
Inner Fan Pressure Ratio	1.1	1.4	1.9
IP Compressor Pressure Ratio	1.5	2	2.5
Outer Fan Pressure Ratio	1.1	1.635	1.9
HP Compressor Pressure Ratio	2	4.5	10
Design Bypass Ratio	8	9.3	12
Intake Pressure Ratio	0.5	0.99	1

Constraints	Min Value	Start Value	Max Value
HPC Exit Temperature T3	550	632.506	750
Overall Pressure Ratio P3/P2	8	12.348	16
Net Thrust	149	150	151
HPT Pressure Ratio	1	1.81476	3
LPT Pressure Ratio	2	4.17185	5
IPT Pressure Ratio	1	1.22058	3

Figure of Merit

○ Maximize ◉ Minimize

Sp. Fuel Consumption

Figure 30: Optimization setup of 3-spool configuration

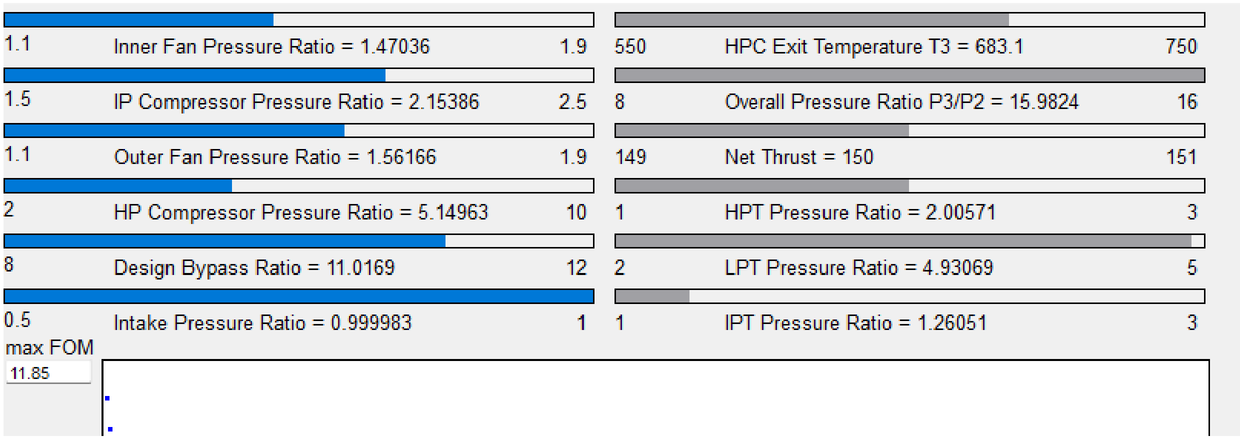

Figure 31: Optimization results for three-spool configuration

Station	W kg/s	T K	P kPa	WRstd kg/s
amb		288.03	100.187	
2	559.392	288.03	100.186	566.836
13	516.360	326.89	150.806	370.305
21	43.032	320.64	141.679	32.533
22	43.032	320.64	141.679	32.533
24	43.032	405.73	298.686	17.359
25	43.032	405.73	292.713	17.713
3	41.526	683.44	1602.540	4.052
31	36.792	683.44	1602.540	
4	38.167	1882.00	1522.413	6.505
41	40.318	1825.44	1522.413	6.768
42	40.318	1588.18	742.787	
43	42.900	1539.47	742.787	
44	42.900	1539.47	736.844	
45	43.761	1522.67	736.844	13.861
46	43.761	1456.84	589.972	
47	43.976	1446.63	589.972	
48	43.976	1446.63	589.972	16.957
49	43.976	1052.34	122.231	
5	44.406	1048.51	122.231	70.360
8	44.406	1048.51	121.009	71.071
18	516.360	326.89	147.036	379.800
Bleed	0.000	683.44	1602.540	

Efficiencies:	isentr	polytr	RNI	P/P
Outer LPC	0.9153	0.9200	0.989	1.505
Inner LPC	0.9160	0.9200	0.989	1.414
IP Compressor	0.8891	0.9000	1.232	2.108
HP Compressor	0.8752	0.9000	1.922	5.475
Burner	0.9995			0.950
HP Turbine	0.8975	0.8900	1.735	2.050
IP Turbine	0.8925	0.8901	1.038	1.249
LP Turbine	0.8969	0.8782	0.882	4.827

HP Spool mech Eff	0.9800	Nom Spd	15764 rpm
IP Spool mech Eff	1.0000	Nom Spd	15000 rpm
LP Spool mech Eff	1.0000	Nom Spd	12800 rpm

hum [%]	war0	FHV	Fuel
66.0	0.00704	43.124	Generic

FN	=	150.00 kN
TSFC	=	9.1641 g/(kN*s)
WF	=	1.37461 kg/s
s NOx	=	0.41655
BPR	=	11.9995
Core Eff	=	0.4246
Prop Eff	=	0.0000
P3/P2	=	15.996
P2/P1	=	0.99999
P22/P21	=	1.00000
P25/P24	=	0.98000
P4/P3	=	0.95000
P44/P43	=	0.99200
P48/P47	=	1.00000
P6/P5	=	0.99000
P16/P13	=	0.97500
P16/P6	=	1.21509
P5/P2	=	1.22004
V18/V8,id	=	0.78348
A8	=	0.40854 m²
A18	=	1.75339 m²
XM8	=	0.54201
XM18	=	0.76137
WBld/W2	=	0.00000
CD8	=	0.93835
CD18	=	0.94874
PWX	=	50.00 kW
WlkLP/W25	=	0.00000
WBld/W25	=	0.00000
Loading	=	100.00 %
e442 th	=	0.87789
WCHN/W25	=	0.05000
WCHR/W25	=	0.06000
WCIN/W25	=	0.02000
WCIR/W25	=	0.00500
WCLR/W25	=	0.01000

Input Data File:

Figure 32: Summary of design point for three shaft configurations after optimization

Observations and Discussion

The SFC as seen decreases with optimization like the 2-spool problem. This indicates that the results are passable and the optimization process was a success.

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